

JAVA MULTIPLE INHERITANCE USING INTERFACES

Object-Oriented Programming Practical Record

1. Aim

To write a Java program using multiple inheritance through interfaces to calculate and display an employee's basic salary, bonus and total salary.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Main.java**.
3. Define the **BasicSalary** interface with the calculateBasic() method.
4. Define the **Bonus** interface with the calculateBonus() method.
5. Create the Employee class implementing both interfaces.
6. Declare salary and bonus values in the Employee class.
7. Implement the methods and calculate the total salary.
8. Create an Employee object in main().
9. Compile and run the program.

3. Algorithm

1. Start the program.
2. Create the BasicSalary interface with calculateBasic().
3. Create the Bonus interface with calculateBonus().
4. Create the Employee class implementing BasicSalary and Bonus.
5. Initialize salary = 30000 and bonus = 5000.
6. Display the basic salary using calculateBasic().
7. Display the bonus using calculateBonus().
8. Calculate and display total salary using totalSalary().
9. Stop the program.

4. Explanation of the Program

BasicSalary interface: It declares the calculateBasic() method for displaying the employee's basic salary.

Bonus interface: It declares the calculateBonus() method for displaying the employee's bonus.

Employee class: It implements both BasicSalary and Bonus interfaces. Thus, Java achieves multiple inheritance through interfaces.

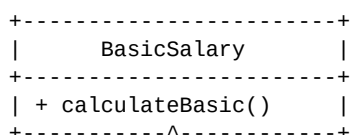
calculateBasic(): Displays the basic salary.

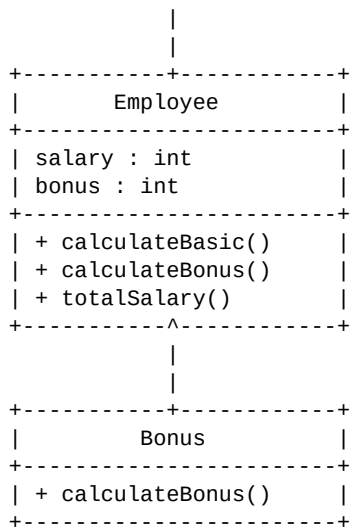
calculateBonus(): Displays the bonus amount.

totalSalary(): Adds salary and bonus and displays the total salary.

Main class: Creates an Employee object and calls all the required methods.

5. UML Class Diagram





6. Java Source Code

```
// Multiple Inheritance using Interfaces
// Real application: Employee Salary

interface BasicSalary {
    void calculateBasic();
}

interface Bonus {
    void calculateBonus();
}

class Employee implements BasicSalary, Bonus {

    int salary = 30000;
    int bonus = 5000;

    public void calculateBasic() {
        System.out.println("Basic Salary = " + salary);
    }

    public void calculateBonus() {
        System.out.println("Bonus = " + bonus);
    }

    void totalSalary() {
        System.out.println("Total Salary = " + (salary + bonus));
    }
}

public class Main {
    public static void main(String[] args) {

        Employee emp = new Employee();

        emp.calculateBasic();
        emp.calculateBonus();
        emp.totalSalary();
    }
}
```

7. Output of the Program

```
Basic Salary = 30000
Bonus = 5000
Total Salary = 35000
```

8. Result

Thus, the Java program successfully demonstrates multiple inheritance using interfaces by implementing BasicSalary and Bonus in the Employee class and displaying the employee's basic salary, bonus and total salary.